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When Every Device Speaks a Different Language

Building Resilient Event-Driven Systems

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Oklahoma Cyber Innovation Institute

THE UNIVERSITY of TULSA

WHERE THIS COMES FROM

OCII Smart City Lab

A smart city scaled down to one lab, so its failures can be studied safely.

WHO

Oklahoma Cyber Innovation Institute

The University of Tulsa

DESIGNATION

NCAE-C

National Centers of Academic Excellence in Cybersecurity

FUNDING

OCAST and GKFF

Grant-funded research

USED FOR

- RF monitoring
- Traffic control
- Healthcare
- Environmental sensing

The story

Two research questions

PROJECT 1 · CITY RADIO LINKS



Collateral damage of counter-drone (C-UAS) measures in urban areas

PROJECT 2 · DRONES OVER OUR CITIES



The risks of drones doing critical missions over a city

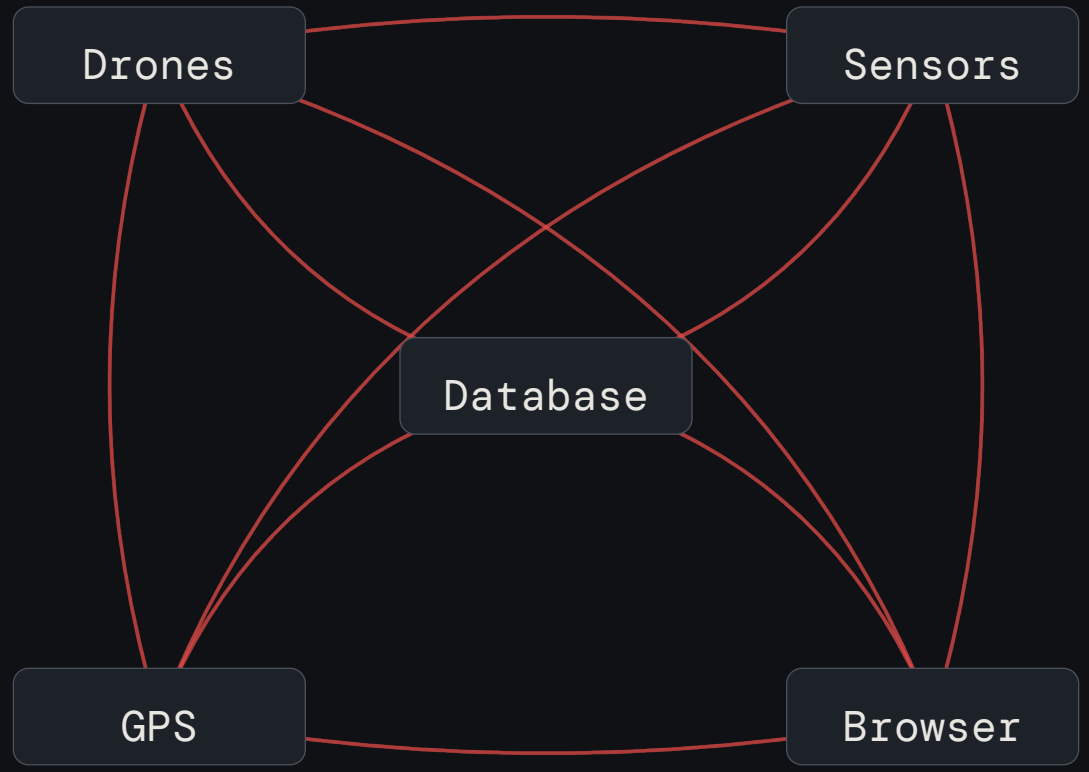
Same Problem Underneath

Dozens of devices, all speaking different languages, and we need to know ***right now*** when one goes quiet or goes wrong. That system is the talk.


KERUI

THE USUAL ANSWER

You build adapters.

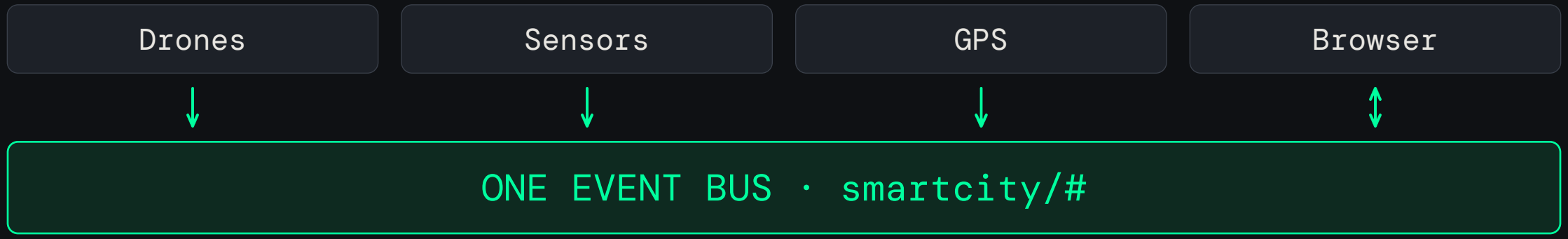


5 systems = 10 adapters.
Add one more: 15.

This is how projects die.

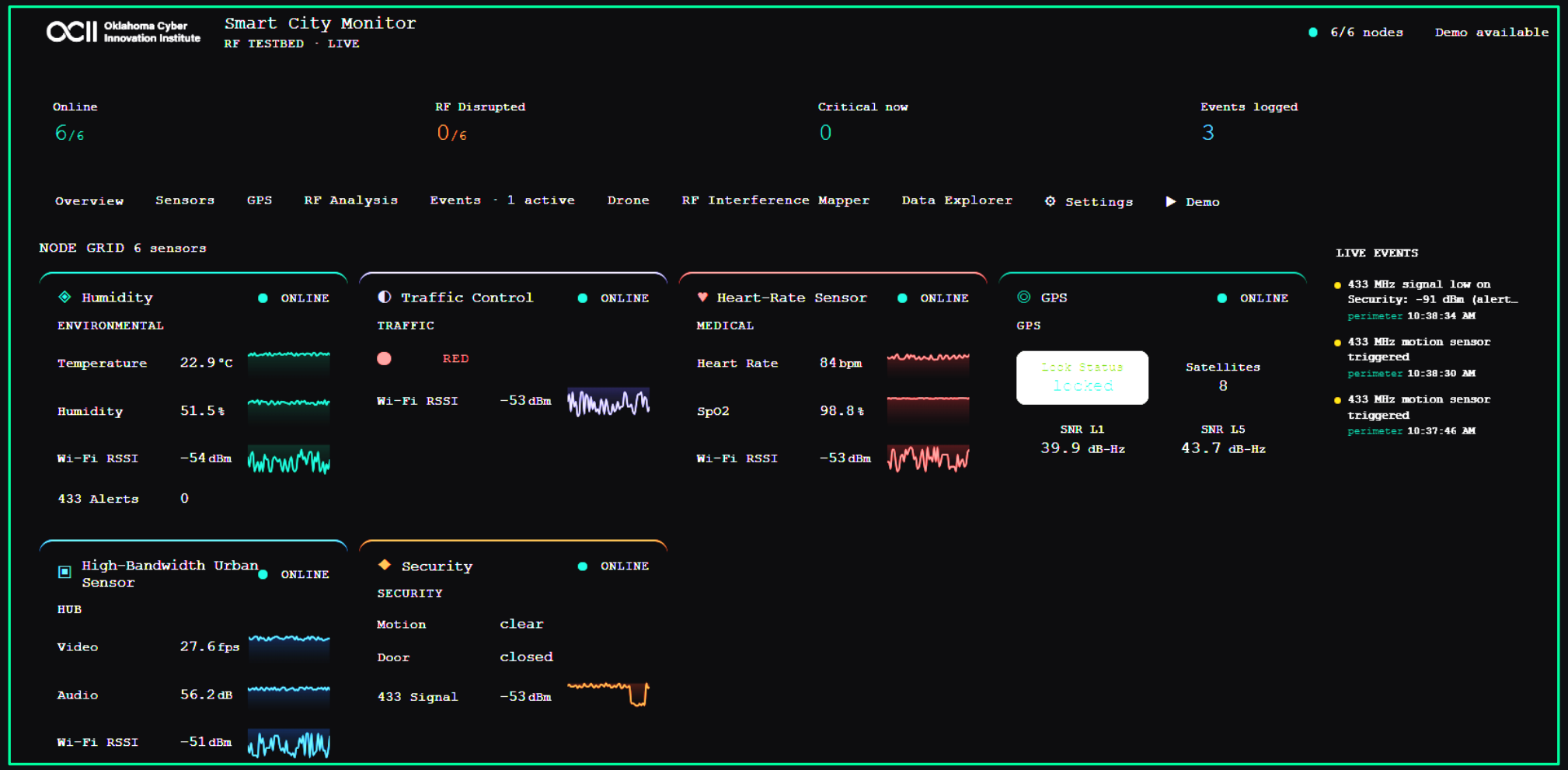
THE ANSWER

One event bus.
Every device speaks to it.
Nothing speaks to each other.



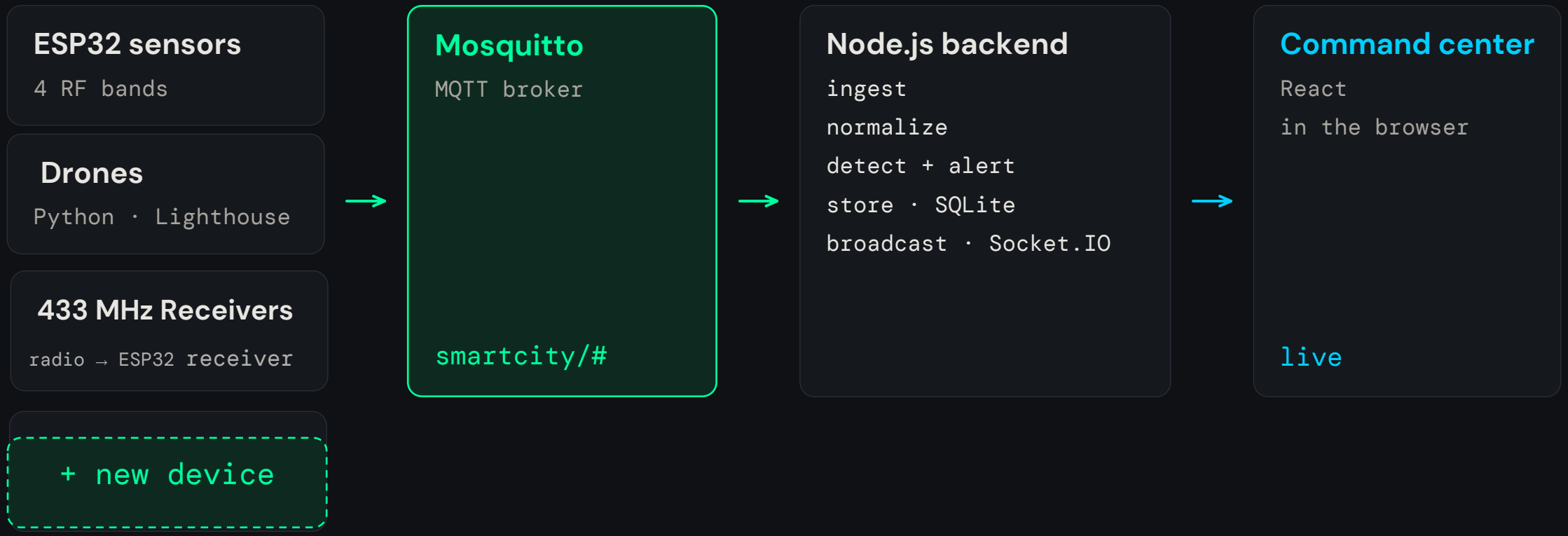
THE COMMAND CENTER

This is what all of it looks like in one place.



THE SOLUTION

Devices → broker → backend → command center



also out: REST API, ROS 2, MCAP recording

THE STACK

The bus is the contract. The language doesn't matter.

C++

ESP32 firmware

Measures, stamps the time, publishes.

MQTT client

PYTHON

Drone services

Fly the drones and track Lighthouse positions.

CRTP · 600 ms telemetry

NODE.JS

Backend

Subscribes once. Normalizes, detects, stores, broadcasts.

MQTT · SQLite · Socket.IO

REACT

Command center

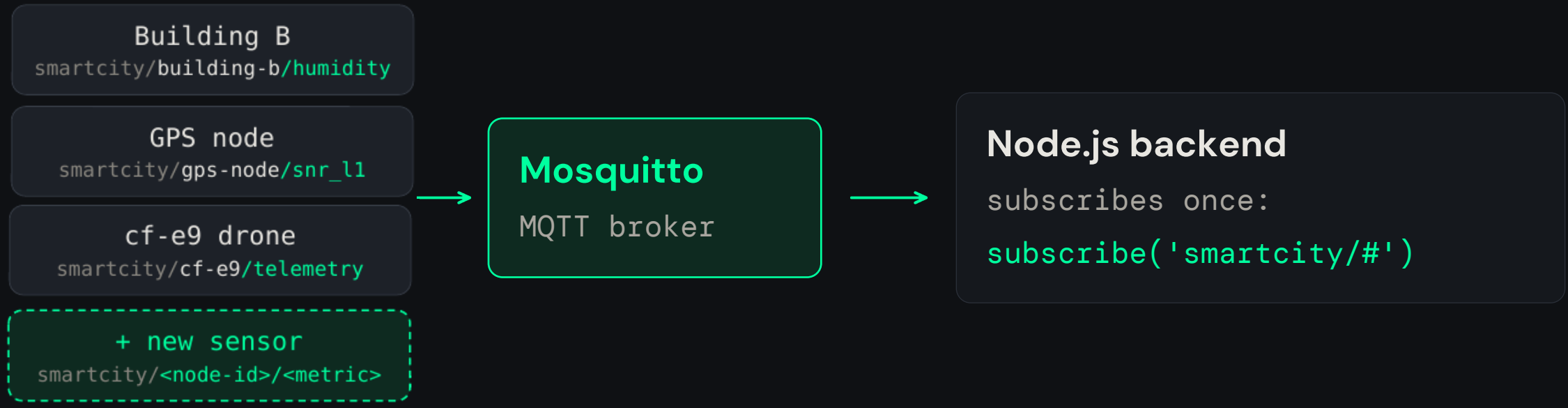
Renders live state in the browser.

Socket.IO client

They only share a topic and message shape. Nothing else.

PATTERN 1 · PUBLISH DON'T CALL

A new sensor just starts publishing. No new code.



DEVELOPERS Services talk through a queue, not to each other.

PATTERN 2 · NORMALIZE AT THE DOOR

Translate once, at the edge.

DRONE
{ x, y, z, battery }

GPS
UBX · B5 62 01 07 …

ESP32 SENSOR
{ temperature, humidity, rssi }



MQTT

EVERY READING, ONE SHAPE (with some tolerance)

topic
smartcity/<node>/<metric>

payload
{ value, unit, collected_ms }

DEVELOPERS Reshape once, where data enters. Never in every consumer.

PATTERN 3 · THE TIMESTAMP IS SACRED

62 seconds of silence arrived in 71 ms.



DEVELOPERS Stamp events where they happen. Arrival time lies under retries and batching.

BEFORE THE DEMOS

Design the system so the hardware is optional.

REAL

Live devices

The podium ESP32 and your phones.

SIMULATED

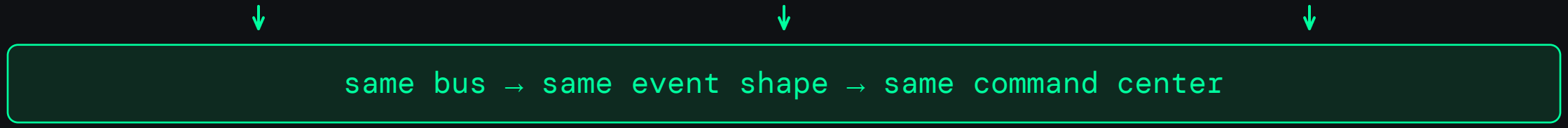
Demo mode

The city's seven sensor nodes.

REAL, RECORDED

Replay

A drone flight from our lab, September 28.



The command center was built before a single sensor was connected.

LIVE DEMO 1 OF 4

RF interference

WHAT YOU JUST SAW

One interference source. 5 devices. 1 incident.

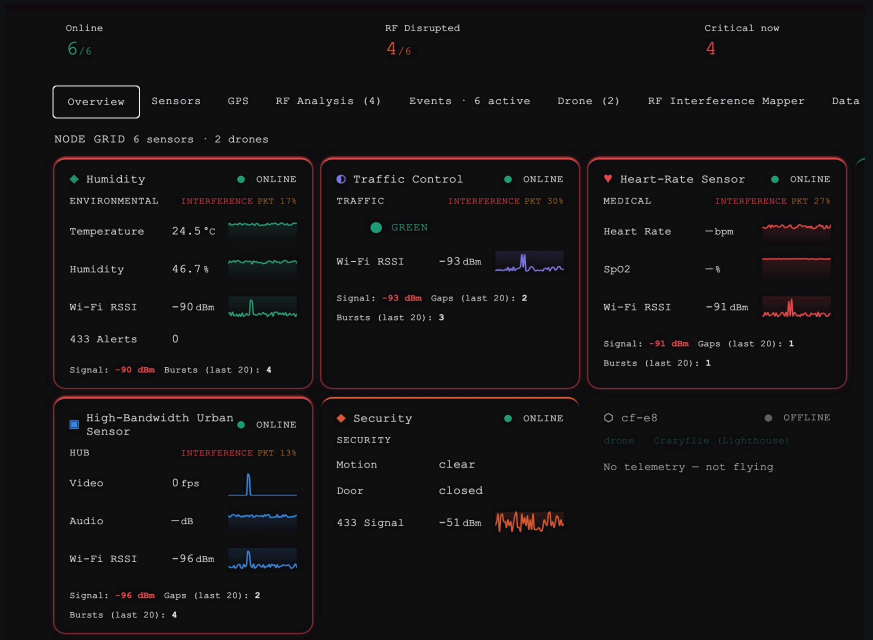
INCIDENT

2.4 GHz interference: 5 devices affected · worst -97 dBm

~300 → 4

events per minute, before

events per cycle, now



Only 2.4 GHz was hit. GPS and 433 MHz stayed healthy.

PATTERN 4 · ONE CAUSE, ONE ALERT

Fire once. Resolve once. Group by root cause.

INCIDENT 2.4 GHz interference: 5 devices affected · worst −97 dBm

pending
The condition has to last before it counts

→

firing
Fires once, then escalates instead of repeating

→

resolved
Clears once, and only past a margin

Filed under it's root cause
5 devices, 1 incident

DEVELOPERS

Page on incidents, not log lines.
ISA-18.2 industrial alarms and Grafana
AWS IoT Events alerting

HOW WE APPLIED IT

Five rules behind every alert

1	**Delays are in seconds, not readings**	`2 s, whether 1 Hz or 10 Hz`
2	**No delay when safety is at stake**	`heart rate · SpO2 · GPS lock`
3	**Recovery needs a margin**	`fires < -85 dBm · clears > -82 dBm`
4	**Offline means “no data”, not “fine”**	`went offline → back online after 6 s`
5	**One cause, one incident**	`held 3 s, grouped by radio band`

DEVELOPERS One config file, thresholds say what’s wrong, policy says when to tell you.

PATTERN 5 · SILENCE IS DATA

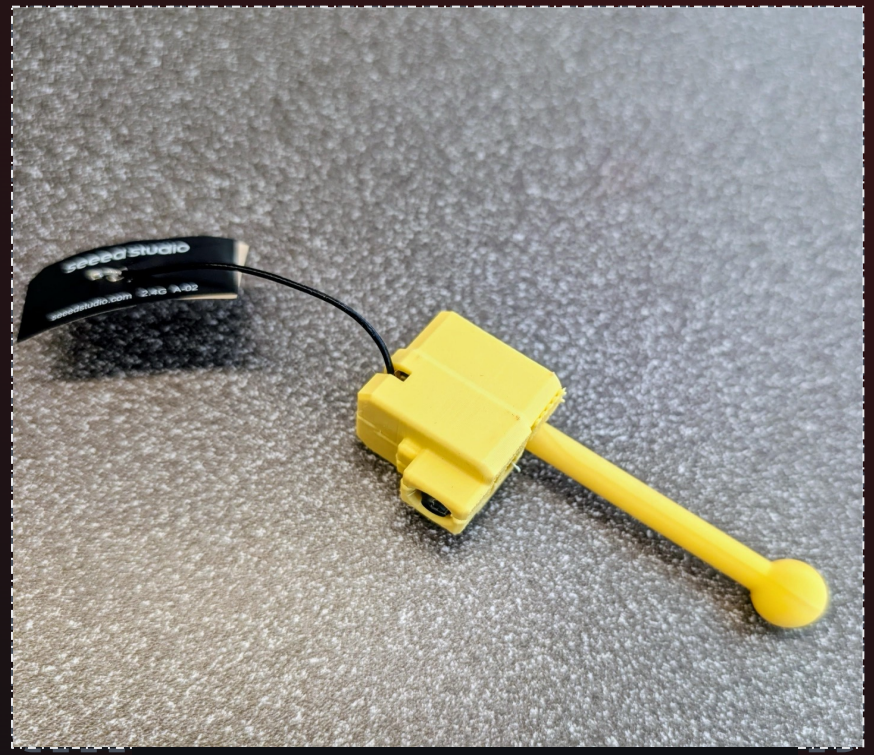
No data for 5 seconds is an event.



DEVELOPERS A heartbeat timeout is an event. Absence of traffic is a signal.

LIVE DEMO 2 OF 4

Unplug a real device



Seeed Studio XIAO ESP32S3 Sense

LIVE DEMO 3 OF 4

Scan to join the testbed

Become a node — ThunderPlains testbed

https://relay-il6i.onrender.com/



You'll see “Hold tight” until joining opens.

DEMO 3 · BECOME A SENSOR NODE

Your phone is a storm

supercell · derecho · wallcloud · mesocyclone

… 18 storm names in all

wallcloud-006

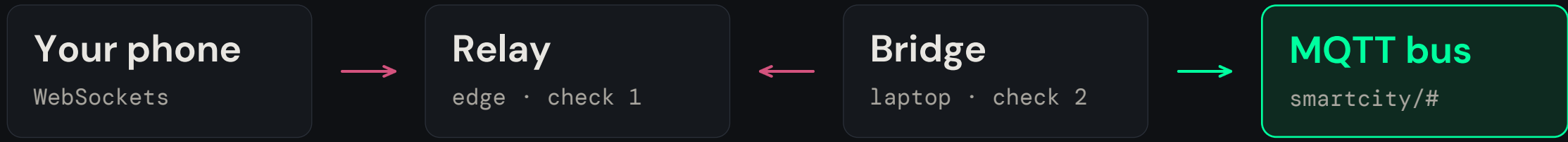
a lowered cloud under a storm, often where tornadoes form



Shake it. Then mash **Ping the stage**.

WHAT JUST HAPPENED

Checked twice before the bus



Nothing you type reaches the screen.

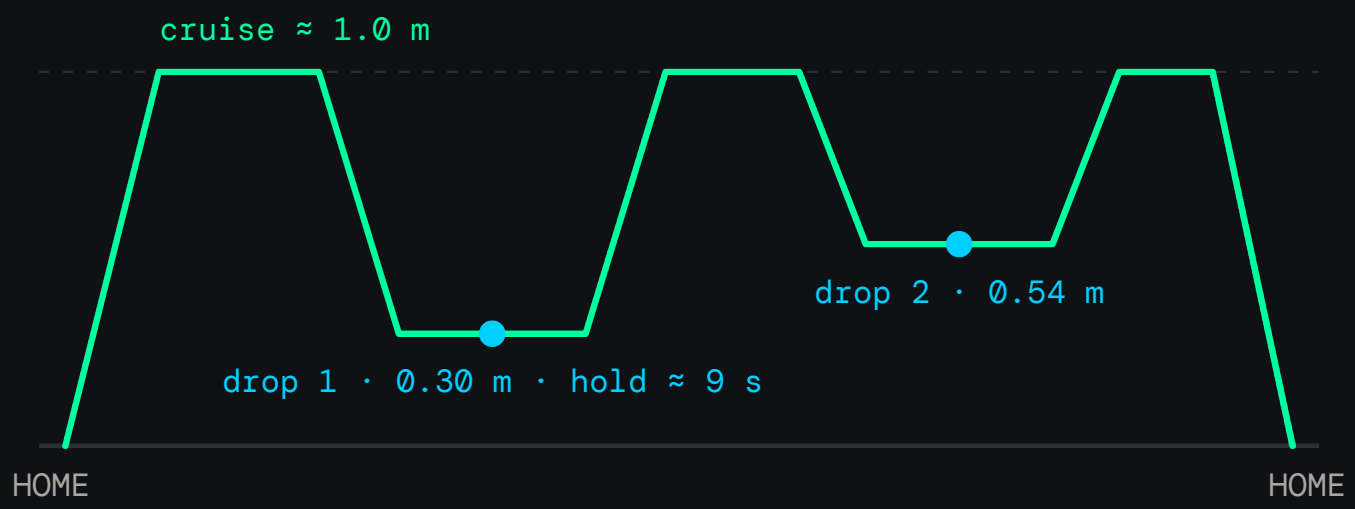
My laptop only dials out.

Close the tab: silence, at scale.

Tested with 290 simulated phones and 5 attacker connections.

DRONES DELIVER PACKAGES

How we simulate package delivery





So the command center has a side view: cruise, descend, hold, climb back.

LIVE DEMO 4 OF 4

Drone attack replay



DEMO 4 · WHAT HAPPENED

Anatomy of the hijack

ATTACK

18s

Attack command sent to cf-e9

→

SILENCE

22.5s

Telemetry stalls right after the command

→

DETECTED

23

Off route, 0.66 m outside its zone

→

Contest

23-24.5

Off route, 0.66 m outside its zone

→

OUTCOME

down

Forced down inside cf-e8's airspace

The limit was the telemetry rate during the attack, that’s not a good detector.

THE RESEARCH · SILENCE IS DATA, AGAIN

What the attack trials showed

50 attack trials

100% Detection

<1.2% false alarms

Random Forest classifier and grouped 5 fold validation method

F1=0.96

“Authority Contention in Micro-UAV Command Stacks” · IEEE CARS 2026

SAME MODEL, DIFFERENT DEVICE

Drones follow the same rules

FIRES ONCE

Off route

Resolves with “back on route,” or closes with the outcome.

`landed off expected path … left its route 9 s earlier`

ESCALATES

Battery

One alert per flight, not a new event at every level.

`monitor → low → critical`

CLOSES ITSELF

Telemetry lost

One alert that ends when the data comes back.

`recovered after 3 s`

The whole drone replay you just watched: 8 events.

THE HARD PART

Here's what actually went wrong.

WAR STORIES · HOW THE PATTERNS WERE LEARNED

Three bugs that taught me the patterns

1	Our dashboard said every sensor was sick. They were fine.	lesson: simulated data is too polite
2	The GPS node was too busy asking to listen	lesson: don’t poll. Let it push.
3	Every clean landing raised “signal lost”	lesson: judge silence against the mission

TAKE HOME

Five patterns, no drones required

1

Publish, don’t call

Nothing talks directly.

2

Normalize at the door

Translate once, at the edge.

3

The timestamp is sacred

Never let the broker assign the time.

4

One cause, one alert

Page on incidents, not log lines.

5

Silence is data

Treat it that way.

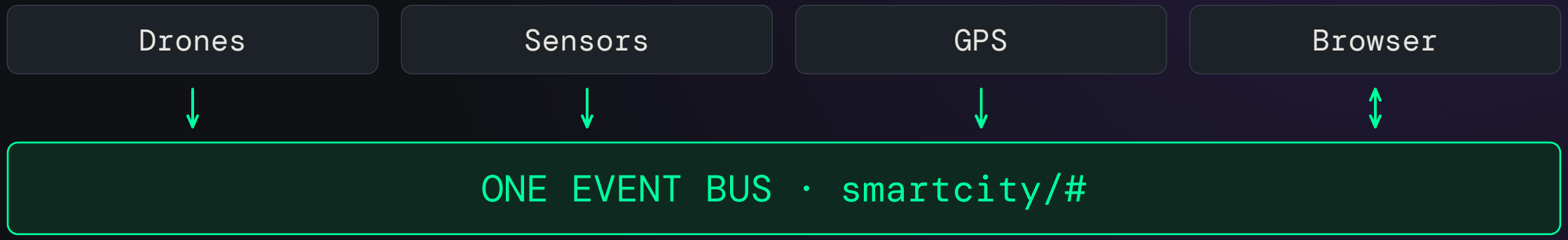
NOT JUST DEVICES

Where else these patterns apply

- Microservices on queues
- Webhooks from many vendors
- CI pipelines
- Apps mixing REST, WebSockets and webhooks

THANK YOU · THUNDERPLAINS 2026

One event bus.
Every device speaks to it.



https://github.com/fatemeh-azam/ThunderPlains2026SmartCityDemo/

Fatemeh Azam · OCII · The University of Tulsa

BACKUP · Q&A

A research testbed with standard alerting

- Alert on state changes, not readings
- Delay before firing
- Wait before resolving
- Avoid alarm floods

THE SAME IDEAS IN

- ISA-18.2 · industrial alarms
- AWS IoT Events
- Azure IoT Central
- Grafana alerting

The standard approach for IoT and industrial monitoring, not only for IT.

BACKUP · Q&A

QoS 0 for sensors, QoS 1 for drones

QoS 0

At most once

Sensors. No acknowledgement and no retry: the cheapest delivery MQTT offers.

QoS 1

At least once

Drones. The broker acknowledges each message, so the sender can retry.

BACKUP · Q&A

How the phone demo stays safe

- Joining is closed until I open it from the Stage tab.
- Every message is checked at the relay, then again at the bridge.
- Nothing an attendee types is rendered on the big screen.
- The laptop only dials out; nothing connects in.
- One message per phone per second.

Tested with 290 simulated phones and 5 attacker connections.

BACKUP · Q&A

SQLite as the event store

WAL MODE

Readers keep reading while the backend writes.

One file, no server to run. Lab build: better-sqlite3. Public repo: Node’s built-in SQLite, nothing to compile.

BACKUP · Q&A

Drone telemetry, ROS 2 and MCAP

TELEMETRY

Every 600 ms

Three log configs, within the Crazyflie's 14-slot limit.

OUTPUT

ROS 2

Drone telemetry is also published for ROS 2 tools.

RECORDING

MCAP

Flights are recorded, and replayed through the same MQTT bus.